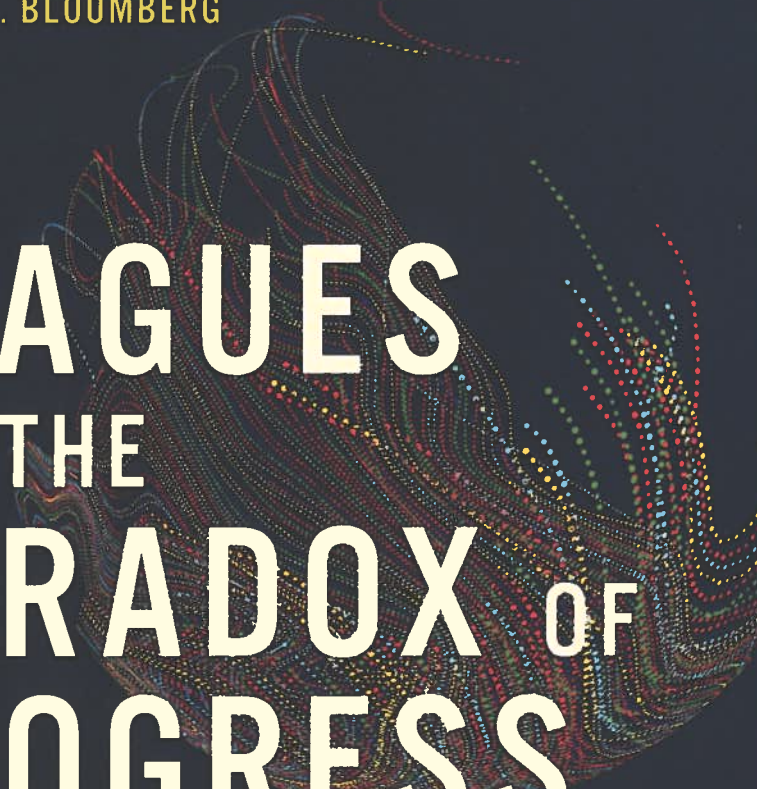


"BOLLYKY'S BOOK CAN HELP TO SAVE MANY LIVES."
—MICHAEL R. BLOOMBERG



PLAGUES AND THE PARADOX OF PROGRESS

WHY THE WORLD IS GETTING HEALTHIER
IN **WORRISOME** WAYS

THOMAS J. BOLLYKY

6 The Exoneration of William H. Stewart

The time has come to close the book on infectious diseases. We have basically wiped out infection in the United States.

—Attributed to William H. Stewart, US Surgeon General, 1967 or 1968

William H. Stewart accomplished a great deal as Surgeon General at a consequential moment in the history of US public health. Stewart, a pediatrician and epidemiologist, held the post of “America’s doctor” when it carried considerably more power than it does now and was responsible for the day-to-day administration of the US Public Health Service.¹ He was appointed in 1965, one year after his predecessor, Luther Terry, had issued a bombshell report linking cigarette smoking to higher rates of lung cancer and other diseases. Stewart managed to get the first health warning labels on cigarette packs, which required facing down the US tobacco industry at the height of its influence and at a time when 42 percent of American adults smoked.² The year 1965 was also when Lyndon Johnson enacted Medicare and Medicaid, which provided health insurance to older and disabled Americans, and one year after the landmark Civil Rights Act. Stewart threatened to withhold federal funds from hospitals that refused to integrate their staff and open their doors to minority patients. He also drew the link between air pollution and lung disease, which helped set the stage for the enactment of the Clean Air Act of 1970. Stewart raged against the “glass curtain” that separated the poor from quality care, built health facilities in underserved areas, and launched a campaign to vaccinate more than 100 million people in Africa against smallpox and measles.³

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the HIV epidemic and has been used in dozens of books, reports, and articles that warn of the perils of complacency on emerging infectious diseases and antibiotic resistance. For the medical community, Stewart's quote has become the equivalent of the advertisements for the "unsinkable" *Titanic*—an apt metaphor for humanity's hubris in grappling with microbes that can never truly be defeated. Stewart died in 2008 and the obituaries for him that appeared in the *New York Times* and the prestigious British medical journal the *Lancet* both cited the statement.⁴

Stewart's line about turning the page on infectious disease has become, according to the *New Yorker* magazine, "one of modern medicine's more famous quotes."⁵ The only problem is that Stewart never said it.

Brad Spellberg and Bonnie Taylor-Blake, physicians and medical researchers, spent years looking for the provenance of the Stewart quote. They worked with the historian for the US Public Health Service and searched every conceivable archive and database. Spellberg and Taylor-Blake traced the quote to a 1989 US National Institute of Allergy and Infectious Diseases (NIAID) meeting on emerging viruses. The HIV epidemic raged unabated at that time and one of the speakers condemned US complacency in preparing for such pandemics. In doing so, he cited Stewart as an example of that complacency, referencing a purported statement that Stewart had made at a 1968 conference claiming that science had probed the frontiers of knowledge about infectious diseases. As it turns out, the speaker had not attended the conference at which Stewart spoke. Several journalists were present at the NIAID meeting and in their subsequent reporting, the reference to Stewart evolved into a direct quote, with some citing Stewart's 1968 speech and others noting an earlier one in 1967. And so, an urban legend was born. Spellberg and Taylor-Blake write that the "spectacularly erroneous quote has been cited innumerable times to underscore ongoing public health problems caused by antibiotic-resistant and emerging infections."⁶ Michael Specter, a well-known science writer, did not originate the quote, but penned an apology to Stewart for having used it in several pieces over the years.⁷

The story of the misquoting of William Stewart is not included in this book to embarrass those who have mistakenly cited the quote over the years. We all work hard to avoid mistakes, but some happen regardless. Indeed, despite my best efforts, there are undoubtedly some even in this book.

Moreover, while those who cited the Stewart quote were wrong, their underlying point is not. In the 1950s and 1960s, there were leading US



Figure 6.1

William H. Stewart speaks at Johns Hopkins University School of Hygiene and Public Health on September 18, 1968. Photo: William C. Hamilton, Courtesy of the Alan Mason Chesney Medical Archives of the Johns Hopkins Medical Institutions.

medical researchers other than Stewart who were skeptical of the continued relevance of infectious disease.⁸ The United States was (and remains) woefully ill-prepared for the threat of antibiotic resistance and future pandemics.

But the Stewart episode is also noteworthy for what he did say in those speeches, not only for what he did not. In that 1967 speech to the Sixty-fifth Annual Meeting of the Association of State and Territorial Health Officers, Stewart emphasized that the "warning flags are still flying in the communicable disease field.... While we are engaged in taking on new duties... we cannot and must not lose sight of our traditional program responsibilities."⁹ Rather than arguing for moving beyond infectious disease, Stewart emphasized the need for continued vigilance against the threat of plagues and parasites even amid changing health needs. That need continues today, and this book should not be taken to suggest otherwise.

The risk of dangerous disease events is ever-present, with the continued emergence of new plagues, such as HIV/AIDS, and the evolution of existing microbes, especially influenza. The risk of such emerging infectious diseases

may be greater than ever with warming temperatures, increased urbanization, the industrialization of agriculture, and higher volumes of trade and travel. Some cancers, such as cervical cancer, are often caused by infectious disease, and many believe there is a link between infection and diabetes. The US Centers for Disease Control and Prevention reported that more than two million people in the United States are infected with antibiotic-resistant bacteria annually, and 23,000 die as a result.¹⁰ An estimated 200,000 people die each year from multidrug-resistant tuberculosis worldwide. Continued global progress against infectious diseases is not inevitable and, without our attention and sustained investment, it could easily be reversed.

But the need for continued vigilance against microbes should not obscure the dramatic progress made in controlling the everyday toll of infectious diseases, nor should it keep us from adapting to the changes brought on by that progress. This is the point that Stewart made in his 1968 speech at Johns Hopkins University, and it is worth quoting at length. Stewart said:

Powerful tools have been developed to characterize these diseases in man and in society—the tools of microbiology, epidemiology, and biostatistics... means for preventing or alleviating disease in an individual or en masse have been developed and applied ... and never before in man's history has a society reached such a peak of health.

This success story brings us to the here and now. And largely as a result of this success story, times have changed. The purpose of our efforts—the preservation and improvement of health—can no longer be measured on the scale of microbiology. Our exploitation of that science has just about caught up with the frontiers of public need. Clearly we cannot turn our backs on microbiology—certain notable gaps remain in our knowledge and capability, and maintenance of a vigilant effort will always be required.

But just as clearly the characterization of health in terms of microbiology and infectious disease epidemiology cannot serve as the base for our future endeavors. Rather, the moving tides of society, which we have helped to move, are compelling us to redefine our purposes in quite different terms—the terms of man's adaptation to his total environment.

Immediately we find ourselves on dark and shaky ground. This new definition of purpose rests on sciences much less comfortably exact than microbiology. We find the familiar equations of one cause-one disease disappearing into a complex of multiple causation.¹¹

This is a fairly accurate characterization of the US public health situation in 1968, the progress that had occurred, and the challenges that lay ahead. As figure 6.2 shows, infectious disease-related deaths in the United States had

declined precipitously in the prior decades (with the significant exception of the 1918 flu epidemic). US deaths from infectious diseases rose again with the HIV epidemic in the first half of the 1990s, but reached nowhere near historical levels. Mortality rates from infectious diseases have fallen since 1995 and reached unprecedented lows in the past two decades. The progress against the noninfectious disease threats that Stewart worried about, such as smoking or pollution, has not been nearly as sharp. So while the threat that pandemic disease and antibiotic resistance pose to the United States continues to loom large, Stewart was right that infectious disease epidemiology should no longer be the exclusive frame through which the United States views its health. During Stewart's tenure, the US Public Health Service launched a major national vaccination campaign against measles, but it also created a system of regional medical centers to tackle stroke, heart disease, and cancer, the diseases that had emerged as the leading killers of Americans.¹²

In recent decades, the same progress that Stewart observed in the United States in controlling plagues and parasites has begun to extend worldwide, even into the poorest regions of the world. Infectious diseases now cause less than one-fifth of the total death and disability in South Asia and less than 44 percent in sub-Saharan Africa. The health transition occurring in many

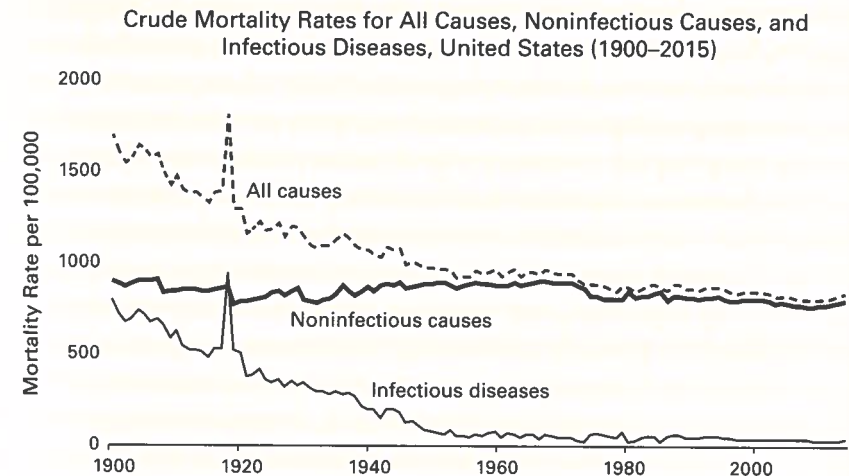


Figure 6.2

Data sources: Armstrong, Conn, and Pinner, "Trends in Infectious Disease Mortality," 1999; Institute for Health Metrics and Evaluation, GBD 2015.

poor nations is much faster than in the past. In 2040, Bangladesh, a country that the World Bank defines as low-income, will have roughly the same share of its health burden dominated by cancer, heart diseases, and other noncommunicable diseases as the United States does today. The difference is that in the United States, that transition occurred over more than a hundred years. In Bangladesh, it will have happened in a third of the time and at a much lower level of income. The health-care systems in many developing nations are completely unprepared for this rapid change. Without access to government-provided chronic and preventative care or the personal wealth to buy it, young, working-age people in poor countries are experiencing staggering increases in noncommunicable illnesses. Four of those diseases—cancer, upper respiratory illnesses, heart disease, and diabetes—already account for nearly six of ten of global deaths annually, yet only 1 percent of development assistance for health is dedicated to noncommunicable diseases.

The speech that Stewart made in 1968 argued for broadening the US approach to health concerns and grappling with the consequences—for good and ill—that had come with the progress made in controlling infectious disease. That argument applies with equal force to many poorer nations now. Dr. Agnes Binagwaho, the former permanent secretary of the Rwandan Ministry of Health, has said: “noncommunicable diseases are often considered the problem of middle- and high-income countries. ... I want to stand strongly against that.”¹³

The rise in these chronic diseases is only one example of the ways that the decline in infectious diseases has, to paraphrase Stewart, moved the tides of society in poorer nations, a move that global health initiatives helped bring about. The implications of that shift extend beyond health, and many of those consequences are profoundly positive. Reductions in infectious diseases have helped raise new economic empires, shift the geography and character of human settlement, and spur the grand movements of people toward a better life. But some of the changes are more worrisome. The growth of many poor world cities is far outpacing their infrastructure, leaving nearly a billion people living in slums. Lack of sufficient jobs for young adults is breeding instability and spurring desperate attempts at migration. Climate change and increased hostility to trade and immigration in the United States and Europe are raising the temperature on a stew of demographic changes that already seems poised to boil over. It is not hard to envision a scenario where the next two decades are even less peaceful than the last.

In short, the history of the decline in infectious diseases has been at least as consequential as the story of their rise. But grappling with the consequences of the partial progress against infectious disease in poorer nations is, as Stewart put it, “dark and shaky ground” that requires confronting “a complex of multiple causation.”

Confronting the Complex of Multiple Causation

Developing countries face multiple challenges: retooling their health-care systems to provide cost-effective preventive and chronic care; improving the reach and sustainability of urban infrastructure; attracting private investment and increasing formal employment opportunities; and extending the progress against infectious diseases, extreme poverty, and undernutrition to those who still suffer from these conditions. High-income countries, including the United States, share these problems, too. The difference is that demographic changes are forcing poorer nations to confront them faster, with fewer resources, amid a changing climate and a more hostile geopolitical environment for trade and immigration.

It is easy to be pessimistic about the prospects for those lower-income countries facing these challenges. But we should not lose sight of the fact that the decline of infectious diseases presents an historic opportunity to invest in the large young adult populations flooding the rapidly growing cities of developing nations as the best hope for generating more inclusive economic prosperity. Cities can be more productive and healthful than rural settings. Urban productivity and healthfulness, if not overwhelmed by traffic and clogged infrastructure, improves with population size and better education.¹⁴ Solutions to these challenges will necessarily be context-specific, but here are three broad areas that developing country governments should consider investing in when facing the paradox of progress against infectious diseases.

First, there is no quick or ready-made fix for creating sustainable urban infrastructure in sprawling poor world cities, but instituting the right incentives can help. Many residents of developing-country cities, especially those living in slums, lack clear property titles to their homes. Establishing easy-to-enforce land rights can promote investment in formal housing, free workers to move to find jobs and get access to city services, and establish the foundation for a property tax system. With the resulting resources and a

bit of empowerment from national governments, city governments may be freer to do more to improve their business environments, ease bureaucratic constraints on small- and medium-sized enterprises, and attract private infrastructure investment. Municipal governments may also prove more capable than national authorities in responding to the demographic pressures and environmental challenges that poor world cities face.¹⁵ A World Bank study of African cities providing the best water services to the poor found mayors behind several of the success stories, including those of Durban, South Africa and Nyeri, Kenya.¹⁶

Second, cities, health-care systems, and economies succeed when their participants have the knowledge and communication skills to pursue and spread new ideas.¹⁷ This requires a good education. Despite dramatic increases in school attendance, many lower-income nations continue to lag far behind their wealthy country counterparts in capabilities such as reading fluently, solving practical problems with arithmetic, and critical reasoning.¹⁸

Lower-income nations, especially in sub-Saharan Africa, are in dire need of more investment in human capital. With declines in infectious diseases and steeply rising rates of child survival, more sub-Saharan Africans will reach the ages of prime employment (ages 15 to 64) by 2035 than will be added by the rest of the world combined.¹⁹ While sub-Saharan Africa has greatly increased school attendance, it is still the worst-performing region in educational outcomes. Up to 40 percent of children in the region do not meet basic learning outcomes in numeracy, and half fall short in literacy.²⁰ In 2016, Ellen Johnson Sirleaf, the president of Liberia, took a radical step to address the problem of failing schools, absentee teachers, and missing books: she delegated management of ninety-three public elementary schools to eight different private contractors. Despite higher costs and more variation in performance across the contractors, the results after one year showed, on average, a 60 percent improvement in educational outcomes in the privately run schools.²¹ More assessment of the Liberian experiment is needed, but the early results are promising.²² Private providers of public education are certainly not the solution everywhere, but they may help improve school attendance and quality in nations with high rates of corruption and poor public services. If so, it will be important to continue to monitor performance and guard against the inequitable access that may arise with privately run schools.²³

Third, a strong primary health-care system can be a cost-effective way to manage many ills, from enabling early identification of a new disease

outbreak to finding a young woman's breast cancer before her tumor is ulcerated and past the stage where it can be treated. But many developing countries have shown too little interest in primary health care. Poorer nations have tended to invest in expensive hospitals that cater to the needs of local elites.²⁴ For example, the *Economist* recently reported that the number of primary-care providers in China declined 6 percent between 2002 and 2013, while hospital capacity almost doubled. The number of hospital beds per person in China now exceeds that in the United States.²⁵

High-quality primary care is not beyond the means of many developing countries; those countries that have instituted it have achieved dramatic health improvements. Brazil's family health program covers more than half of the population, costs the government roughly \$50 per person, and has sharply reduced the number of deaths from heart disease, diabetes, and infectious diseases.²⁶ Community health worker programs that use low-cost, simple diagnostics and digital technologies can extend basic health care to the poor in rural areas and slums.²⁷ Developing countries that have invested in their own capacity to generate and apply evidence-based, localized health policies, like Mexico, have made the greatest gains in public health.²⁸

The Role of Aid in Adapting to the Decline of Infectious Diseases

Determining a country's priorities and allocating scarce resources in the face of its demographic challenges are ultimately the responsibility of national governments and local civil society, which have the most at stake in the future of their own nation. Aid represents a small and diminishing share of overall investment to lower-income countries. Moreover, aid agencies and other donors are also outsiders, poorly positioned to undertake the sustained, hard struggle of adapting institutional and social reforms to local needs and advancing them on behalf of lower-income countries.

Past foreign aid initiatives succeeded in delivering food, cash, drugs, and other technologies to the world's poor, even in countries with dysfunctional governments. But donors and international agencies have a dismal record of advancing the political and social reforms that would help poorer nations seize the potential of their improved health. Donors cannot manage primary health-care systems in other countries or enforce sensible regulations on tobacco, land use, or air pollution. Aid agencies and philanthropists

have had little success creating economic growth and employment in other nations, let alone installing democratic and responsive governance or sustainable urban planning. It is hard for bilateral donors, reliant on the opinions of their domestic constituents, to justify investments in infrastructure that local governments are ultimately responsible for providing: roads, bridges, sewers, and hospitals. It is the height of hubris to believe that philanthropy and foreign aid can stop the social, political, and economic forces that lead young adults to aspire to a better life and emigrate.

But that doesn't mean foreign aid officials, philanthropists, policymakers, and other readers of this book cannot assist lower-income nations in their difficult tasks. Progress will depend on mobilizing more support for changing the way international assistance seeks to advance health and development in poorer nations.

In Stewart's 1968 speech, he spoke of the need for humankind to adapt to the changes in the total environment, brought on in part by the decline in infectious diseases. Below are three critical points that should serve as benchmarks for assessing international initiatives and their engagement on that agenda.

First, if international development initiatives are going to remain effective as a means of advancing health and economic development, they must be more responsive to the changing needs of lower-income nations. This may sound like an obvious point. But it requires making existing aid and health programs less focused on donor-directed inputs—specific disease-reduction targets, years of primary schooling, and “dollar a day” poverty—and more concerned with local outcomes such as learning, capable economies and governments, and better health, especially among the poor and disenfranchised. At a 2013 conference dedicated to ending child mortality globally, Richard Cash, the co-inventor of the oral rehydration solution that saved so many lives, now a professor at the Harvard School of Public Health, said:

By simply focusing on mortality, which so much of the focus has been on, we ignore issues such as contaminated water, such as poor educational systems... that other package of development issues which are critical. What we don't want is successes in mortality and people still living in abject poverty, and in fact that is what has happened. Our tools are very powerful, our vaccines are very good, our interventions are really good, but we haven't adequately addressed the underlying risks and causes of both the mortality and the morbidity.²⁹

Global health priorities must also include the noncommunicable diseases and associated health risks that now cause the largest proportion of

premature death and disability in lower-income nations. It is simply not sustainable to commit substantial resources to fighting one set of treatable and preventable diseases only to watch the very same patients die prematurely as a result of equally treatable and preventable conditions. The 2017 launch of Resolve to Save Lives, a \$225 million initiative that seeks to improve hypertension control in lower-income countries, provides hope for more donor interest in the rising tide of noncommunicable disease.³⁰ The extensive platforms that the US government and other donors have built to combat HIV/AIDS and other infectious diseases also provide a positive legacy. For example, most international infectious disease initiatives have facilities to ensure the provision of affordable, quality-assured medicines; allowing lower-income countries to use those facilities to procure essential drugs for cancer, diabetes, or noncommunicable diseases would require the assent of donors like the United States, the United Kingdom, and the Bill and Melinda Gates Foundation. Providing that support for expanding those global programs that provide lifesaving commodities and services will be important, particularly for the poorest slums and most conflict-ridden states that face a crippling dual challenge of infectious and noncommunicable diseases.

The United States and other donors should recast their global health aid programs from disease-focused goals to more outcome-oriented measures for improving health in the targeted countries and populations. Adopting a patient-centered approach based on health outcomes rather than disease-specific reduction targets, and investing in data collection to monitor it, would increase the accountability and capability of global health investments. That would make those global health programs more valuable and easier for governments of lower-income nations to move from being largely dependent on outside aid to operating the health programs themselves. Ensuring that developing countries have greater say in choosing the outcome measures chosen will improve their utility and increase the likelihood of their eventual local adoption.

Central to understanding how to advance the interests of those in need is listening to what they actually want. In 2015, Angus Deaton and Robert Tortora analyzed data from multiple polls averaged across sub-Saharan Africa and found that respondents prioritize a better livelihood—new jobs and agricultural improvements—over gains in health care. This does not mean aid initiatives should no longer prioritize health, where they have been most effective. Evidence suggests respondents' desire for additional health aid may have declined in part because infectious disease has lessened

and their greater satisfaction with the existing aid programs.³¹ It does suggest, however, the need to pair future health aid with more investment in other areas that may also help firms and factories, large and small, employ people productively.

Education is another example of where an outcome-based approach may help. Lant Pritchett, in his book *The Rebirth of Education*, argues that the dogged pursuit of the global goal of universal primary school attendance has obscured the lack of achievement in the underlying objective of education: learning. The United Nations Sustainable Development Goals, released in 2015, include objectives on educational quality, but there is no agreement on how to measure progress. Many countries resist being compared to their peers and there are often wide discrepancies between official literacy rates and those reported in surveys.³² In 2017, the World Bank announced an initiative that may help. It will compile a Human Capital Project, an annual evidence-based and transparent assessment that will combine measures of health outcomes and educational quality along with a country's gross domestic productivity. The idea is to encourage private sector firms to use this index in their foreign investment decisions, thereby creating a strong financial incentive for governments to devote more resources to better education and health.

Thoughtful and targeted investments can also help lower-income nations confront the demographic challenges that come with the decline in infectious diseases. In recent years, donors have increased investments in voluntary family planning and girls' education. These programs have helped to reduce fertility in nations like Senegal to more sustainable levels and better integrate women into the economy. These investments should be increased. Recent efforts to encourage private investors to put their money toward building infrastructure and electricity generation, such as the Obama administration's Power Africa initiative, can make it easier for factories and entrepreneurs to open businesses and hire more young workers.

The second step on the road to a less worrisome future is for philanthropists and foreign aid agencies to embrace a more limited but still useful role in helping poor and emerging economies confront the demographic challenges that have accompanied declining rates of infectious disease. This role should focus on funding the data collection and research that advocates and activists need to advance locally inspired change. This effort should include providing the resources and technical support that local

researchers and governments need to test their best ideas and working with those local actors to reduce the results to implementable, evidence-based programs. Outsiders can, where appropriate, also provide a useful service by supporting social reformers in pressing their governments for the adoption of the better policies and by monitoring, evaluating, and publishing the results to hold local governments and donors accountable.

Bloomberg Philanthropies, a philanthropic institution that supports some of my research, has used this strategy to help poor and emerging economies advance tobacco control.³³ In 2008, for example, Turkey raised cigarette taxes to account for 81 percent of the total price of a pack and banned tobacco advertising and smoking in public places. The next year, admissions for smoking-related disease to hospital emergency rooms in Turkey fell by nearly a quarter; smoking rates declined 16 percent over three years.³⁴ The Philippines has also recently adopted similar reforms.³⁵ This data-supported, locally driven strategy on tobacco control has worked in poor and wealthy countries alike and in the face of fierce opposition from the multi-billion-dollar cigarette industry. There is no reason it cannot be used to help advance reform on other issues: more capable health-care systems, family planning, and higher-quality education for girls.

Third, politicians in the United States, Europe, and other wealthy nations must come to terms with the inconsistencies in their policies on climate change, global health, trade, and immigration. Failure to face the reality of climate change and its causes is hurting the economies of many lower-income countries, threatens to undermine progress on global health, and may accelerate the rates of international migration. Researchers have identified, for example, a link between the effects of climate change (declining rainfall and warming temperatures) and childhood stunting in Kenya since 1975.³⁶ Many of the fast-expanding cities in lower-income countries are on the coast or downstream from great rivers. This puts Guayaquil, Ecuador; Ho Chi Minh City, Vietnam; and Khulna, Bangladesh, for example, among the cities that are most vulnerable to floods caused by global warming.³⁷

Politicians in the United States and Europe are increasingly campaigning for office—and sometimes winning—on a policy of anti-immigration and economic nationalism. A mandate from voters should be respected, but wiser politicians and policymakers, rather than exploit the fears of their constituents, would also recognize and explain to voters that those policies work at cross-purposes. Protectionism undermines economic growth and

job creation in emerging economies. In the near term, rising incomes and employment opportunities alone will not stop immigrants from countries with recent declines in infectious disease, but they may shorten the duration and intensity of that migration. Voters in Europe and the United States can choose to be antitrade or anti-immigration, but it is hard to achieve both goals at the same time.

In a globalizing world, little stays local for long. Increasingly, what happens within the borders of any country can affect many others. This is certainly true for the demographic challenges spurred by the partial success in controlling infectious disease. Richard Haass, the president of the Council on Foreign Relations, has called for greater adoption of the notion of sovereign obligation, in which governments have obligations to control the risks and policies in their territory that could adversely affect others.³⁸ It is a doctrine of hard-nosed self-interest and necessity, and an idea that applies with equal force in this context. Lower-income nations must move quickly to adapt to their demographic changes and seize the opportunities offered by their improving health. But wealthy governments must also recognize their obligation to do no harm to those efforts with shortsighted positions on climate change and overly restrictive policies on trade and immigration. One of the best ways to assist poorer nations is to stop obstructing them from undertaking appropriate measures to help themselves. The notion of sovereign obligation is a recipe for intergovernmental cooperation in a world where the day-to-day threat of infectious diseases is diminishing. It is an idea that, together with the other recommendations advocated here, could help make the world not only better, but also less worrisome.

The Myth of the Good Epidemic

The historian Christopher Hamlin has cautioned against the "myth of the good epidemic," the notion that new outbreaks of cholera, tuberculosis, and other infectious scourges might motivate needed investment in sanitation and other positive government reforms, as some argue occurred in the past.³⁹ I also do not subscribe to that myth. The lessons of the past are not that controlling infectious disease and improving the health of the poorest are too much trouble or that progress against pestilence came prematurely to developing nations. There is no worthier goal than reducing unnecessary pain and early death, especially among children.

At the same time, many governments and nonprofits working to promote better health in poorer nations fail to consider that declining mortality rates from infectious disease—the goal that many count as their primary index of progress—will also bring profound challenges if not paired with broader gains in economic development, governance, education, and infrastructure. It is only through an improved understanding of the difficulties in the path that has been taken to reduce infectious diseases that these future obstacles can be overcome, forewarned with a more realistic sense of the deep challenges that lie ahead.